ESYS 104: The Earth System
Assignment #1

Due **September 28, 2026**. Submit on myCourses 'Assignments'. Please don't submit photos of handwritten answers -use a text editor. Assignments turned in late will lose 10% for each day or fraction of a day that you are late. You may work together, but **your answers must be unique and written in your own words**. LLM (ChatGPT, etc.) answers are **not** your own words.

Question 1: Plate Tectonics

Go to: https://dinosaurpictures.org/ancient-earth
Enter your home city, and see where you were on the globe, if you were alive: 220 million years ago (the start of the Age of Dinosaurs) & 470 million years ago (the age of the rocks (mostly) on the island of Montreal). **Give the location, and: Were you above sea level? Near an active plate boundary?**

Watch the two videos below from the University of Texas, Dallas Geoscience Department and answer the questions. Note that these videos are aimed at Geology majors, so some of the concepts and terminology will be foreign to you. That's OK -you can still get something out of them!

Science Behind Mexico's 2020 Earthquake
https://www.youtube.com/watch?v=mIlQqfj8MQY
a) What kind of faulting occurs in along the subduction zone? At what depth did this EQ occur? What are two other large subduction interface earthquakes mentioned in the video?
b) Does Canada have a similar plate boundary to that in Oaxaca? If so where? Which cities are vulnerable?

What caused Iceland's volcanic eruption?
https://www.youtube.com/watch?v=7ljWi-4syBI
a) What is the source of Iceland's volcanism? Which kind of plate boundary is it part of? Which two plates?
b) What are two reasons that people can sit and watch the recent Iceland eruptions rather than run for their lives? What's different about this magma compared to, say, Mount Vesuvius (Italy), Mt St. Helens (Washington State, USA), or another subduction zone volcano?

Question 2: Materials and Resources

a) What is the composition of the Earth? How is this different from the average composition of the Earth materials that you are likely to find on the Earth's surface? List the five most abundant elements in the Earth and the surface layer. What is the surface layer of the solid Earth called? (short answer)

b) In class and in the readings, we explored the formation of three materials that are economically important to Quebec: iron ore (banded iron formation); aluminum (aluminum ore is called bauxite); and diamonds. What are the three geological processes that concentrate these materials into deposits that can be mined (one sentence each answer)?

c) Aluminum ore is not formed in Quebec (why not?) but aluminum ore is smelted into aluminum in Quebec. Why? (short answer; two sentences max)

d) The ocean contains hundreds of tonnes of gold dissolved in the water. All you have to do to be immensely rich is to filter that gold out of the seawater and sell it. Why is this not done? It's the essence of the distinction between the terms 'Reserve' and 'Resource' of an ore. (short answer; two sentences max)

Question 3: Energy Budget (we'll go over this in class on Wednesday)

i) Calculate (show your work) the Radiative Equilibrium Temperature of the pre-industrial Earth, considering:

- The Earth to be a black body
- The atmosphere to be a gray body with emissivity of 0.8
- The planetary albedo to be equal to 0.3.
- The absorptivity of the atmosphere to solar radiation (a) is equal to zero. ie. the atmosphere is transparent to solar radiation.

ii) It has been proposed to seed clouds in the stratosphere, which would increase the planetary albedo of the Earth and regulate the global mean temperature in spite of rising CO_2 levels in the atmosphere.

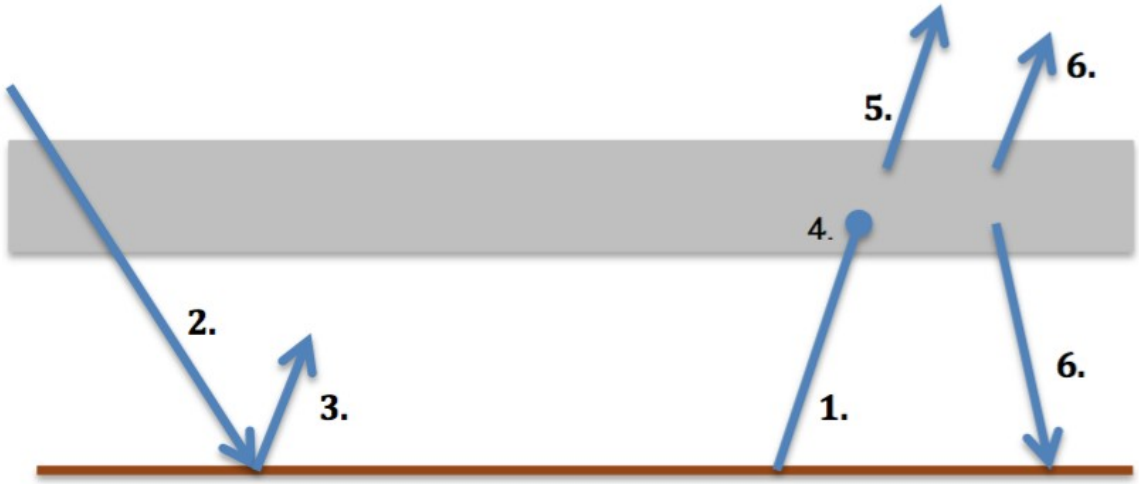
Consider a doubling of CO_2 concentration in the atmosphere from pre-industrial level (280 ppm) to 560 ppm (perhaps around year 2050) leads to a new atmospheric emissivity of 0.83. By how much would we need to increase the planetary albedo in 2050 to maintain the same global mean temperature as in the pre-industrial time?

iii) If clouds are 100% reflective (an approximation; an albedo of 1), then what area of the Earth does this increase in albedo correspond to if it was clouds? Compare this to the size of Canada.

Surface area of the Earth: 510,072,000 square kilometers

Surface area of Canada: 9,984,670 square kilometers

Here's some terms defined (taken from the lectures of the week of 19 January):
Temperature is the balance between energy in and energy out. The Earth's surface is heated both by the sun (shortwave radiation) and the atmosphere (re-emitted longwave radiation). So: a picture and some definitions:



T_e Temperature of the Earth's surface
T_a Temperature of the top of the atmosphere
S_0 The solar constant at the radius of Earth's orbit
α Earth's albedo (amount of the Sun's energy reflected back into space)
ε Emissivity (0-1) is thermal radiation absorbed or emitted. For a true blackbody ε =1
σ Stefan-Boltzmann constant
We're assuming that the atmosphere does not absorb shortwave radiation

1) Earth's surface outgoing energy is σT_e^4
2) The incoming solar radiation is $S_0 / 4$

3) The Earth's albedo, α, reflects 30% of incoming solar radiation back into space.

4) The atmosphere absorbs, at a rate of ε_a, some of the longwave radiation emitted by the surface. The energy it absorbs is $\varepsilon_a \sigma T_e^4$
5) The rest of the longwave radiation emitted by the earth passes through and out into space. This energy is $(1 - \varepsilon_a)\sigma T_e^4$
6) The energy the atmosphere absorbed it reemitted both into space and back towards to earth. This energy is $\varepsilon_a \sigma T_a^4$
From this information, we can set up three energy balance equations. One for the top of the atmosphere (TOA), one for the atmospheric itself, and one for the Earth's surface. For each case, the incoming energy must balance the outgoing energy. There are two unknowns: T_a and T_e., so we actually only need two of the three.

Here's one energy balance equation, to get you started: Earth's surface:
incoming solar + incoming atmospheric emission = earth emitted longwave
$S_0/4\ (1-\alpha) + \varepsilon_a \sigma T_a^4 = \sigma T_e^4$